Danya Mahmoud Saleh Musbah

Software Engineering Graduate | Applied AI, NLP & Data-Driven Systems

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PROFESSIONAL PROFILE

Software Engineering graduate of the University of Tripoli (overall grade Excellent, 92.78%), ranked first among all graduates of the Fall 2025 semester and recommended for postgraduate admission by two faculty members. Built AI-enabled systems end to end: a retrieval-augmented assistant over a curated legal corpus, an Arabic job-posting extraction and classification pipeline validated through manual error analysis, and an LLM- and embedding-based meeting assistant in an individual graduation project. Strong database foundation (100/100 in both database courses) and algorithms (96). Machine-learning tools (TensorFlow, scikit-learn) used in coursework and guided projects. Seeking a fully funded Master's programme in Artificial Intelligence or Data Science to build rigorous foundations in machine learning and research methods.

EDUCATION

Bachelor's Degree in Information Technology, Major in Software Engineering — **Completed Fall 2025**

Faculty of Information Technology, University of Tripoli, Libya

- Overall grade: **Excellent (92.78%)**. Ranked first among all graduates of the Fall 2025 semester.
- Selected coursework (marks out of 100): Introduction to Databases (100), Advanced Databases (100), Design and Analysis of Algorithms (96), Numerical Analysis (96), Data Structures (93).
- Graduation project: *Junto – Real-Time Smart Virtual Collaboration Platform* (individual), grade 94%. Supervisor: Dr. Mohamed E. Gumah Abdulrahman.

Higher Diploma in Pharmacy — **2021 – 2023**

Tripoli Higher Institute of Medical Sciences, Tajoura, Libya · Two-year programme · GPA 86%

HONORS & ACADEMIC RECOGNITION

- **Certificate of Appreciation for Academic Excellence**, Faculty of Information Technology, University of Tripoli (7 May 2026): awarded by the Faculty Council for outstanding academic achievement in the Fall semester of the 2025/2026 academic year; presented at a recognition ceremony with the University President and the Dean.
- **Graduation project exhibition:** presented Junto at the University of Tripoli's academic exhibition.
- **Faculty recommendation letters** (June 2026) from the Head of the Software Engineering Department and a graduation-project examiner recommend postgraduate admission; one also recommends a Teaching Assistant appointment. Available on request.

ACADEMIC RESEARCH

The Effects of Levothyroxine on Bone Density — **2023**

Academic research completed as part of the Higher Diploma in Pharmacy (unpublished)

PROFESSIONAL EXPERIENCE

Data Entry Operator & System Administrator — **Sep 2026 – Present**

Pioneers of Science School, Libya

- Enter student data into the Ministry of Education's Al-Shati system, using the school account set up with the access code issued by the Ministry.
- Also handle IT troubleshooting for the school (276 students, 96 employees), and prepare reports, official correspondence, formatted documents, and design materials.
- Propose process improvements to school management and take responsibility for implementing them.

SELECTED SOFTWARE PROJECTS

Animal Rights AI – Evidence-Grounded RAG Assistant — **Individual Project**

AI-assisted development

- Designed the architecture and technical decisions for a retrieval-augmented assistant over a curated animal-welfare law corpus (FastAPI, PostgreSQL/pgvector, Supabase), specifying hybrid semantic and full-text retrieval with Reciprocal Rank Fusion, source citations, and an insufficient-evidence response.
- Implemented with AI coding tools and reviewed by me; includes a pytest suite and a small evaluation question set (no benchmark results reported).

Libya Jobs Intelligence – Labor Market Data Platform — **Individual Project**

AI-assisted development

- Designed an automated pipeline that collects real job advertisements from Telegram channels, normalizes Arabic text, classifies posts (JOB / NON_JOB / UNCERTAIN), extracts structured fields, and detects duplicates; PostgreSQL/Supabase schema separating original posts from normalized job records, with an Arabic-first website for job discovery and analytics.
- Wrote the initial extraction logic, then manually compared extracted fields against each original Telegram post, identified recurring errors, and used GitHub Copilot to implement fixes across 30+ refinement cycles.
- Python, Telethon, PostgreSQL, Supabase, GitHub Actions, Pytest, JavaScript.

Junto – Real-Time Smart Virtual Collaboration Platform **2025 – 2026**

Individual Bachelor's graduation project, University of Tripoli · Grade 94%

- Independently carried out requirements analysis, design, database, frontend, backend, AI integration, and testing/documentation for a bilingual (Arabic RTL / English LTR) platform with workspaces, WebRTC meeting rooms, real-time chat, shared calendar, file sharing, and AI-assisted search and meeting features.
- Designed an 88-table PostgreSQL database (ERD documented in the repository) and implemented JWT-based authentication with OAuth login (Google, GitHub, Discord) and active-device management.
- Documentation includes 102 functional requirements, an ERD, use-case diagrams, 20 activity diagrams, and 17 sequence diagrams.
- Built with Node.js, Express.js, PostgreSQL, Drizzle ORM, Socket.io, WebRTC/PeerJS, Gemini API, and sentence-transformer embeddings for semantic search (RAG). Designed with a lightweight frontend for unstable connectivity (design consideration; not benchmarked).
- Demo: youtu.be/K1SkB1EjjYQ · Code: github.com/danya-musbah/Junto

University Team Projects **University of Tripoli**

- *Java Bike Store* (Semester 3, team of 3, team lead): built the project foundation, then assigned tasks; developed the backend and system integration using Java and Microsoft SQL Server.
- *Wonder Libya* (Semester 4): responsible for design diagrams, UI design, and the Software Design Document; also designed and coded the booking management feature (create, view, edit, delete).
- *Electrohub Online Store* (Semester 5, team of 6, team lead): managed task distribution and covered gaps in the team. Personally designed the MySQL database (about 30 tables) and built the Laravel/PHP backend with role-based access (guest, user, admin) and protection against SQL injection, XSS, and CSRF.

VOLUNTEER SOFTWARE DEVELOPMENT EXPERIENCE

Volunteer Software Developer **Ongoing**

Pioneers of Science School, Libya · unpaid, outside paid duties

- Independently designed, built, deployed, and documented a school management system covering student and guardian records, academic years and classes, timetables, exams and results, attendance, staff records, fees, reports, and documents. Requirements were gathered through meetings with the principal and deputy principal and interviews with each department's staff, from finance to student affairs. Not every module is fully production-verified.
- Built a student portal (profile, attendance, approved results, fees, timetable) used by students; issued credentials in classrooms and taught students to use it. Trained staff on the management system.
- Stack: HTML, CSS, JavaScript modules, Supabase (Auth, PostgreSQL, RLS, RPC, Edge Functions, private storage), Cloudflare Pages.

TECHNICAL SKILLS

Data science: NumPy, pandas
Programming languages (applied in projects): Python, JavaScript, TypeScript, PHP, Java, SQL, HTML5, CSS3
Databases: PostgreSQL, Microsoft SQL Server, MySQL, Supabase
Real-time & AI integration: Socket.io, WebRTC, RAG and semantic search, LLM APIs
Backend & web frameworks: Node.js, Express.js, FastAPI, Laravel, EJS, Chart.js, JWT authentication
Coursework and guided projects: C, C#, C++ (Unity), JavaFX, TensorFlow, scikit-learn
Cloud, deployment & tools: Cloudflare Pages, Vercel, Render, GitHub Actions, AWS, Azure, Google Cloud, Git/GitHub, ESLint, Figma
Applied learning: *100 Days, 100 Projects* (ongoing): guided implementations extended with my own modifications.

LANGUAGES & INTERESTS

Arabic: Native · **English:** Upper-intermediate, self-assessed (approx. IELTS 6.0); official test result to be provided if awarded.
Areas of interest: applied AI, natural language processing (Arabic), retrieval-augmented generation, data engineering and analysis.